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AUTOMATIC IMAGE CONTRAST ENHANCEMENT BASED ON REINFROCEMENT LEARNING

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Abstract:

Image contrast enhancement is a subjective problem depending on personal preference and subject field property. Every person has different perception on the assessment of an enhanced image quality. Thus, it is difficult to have one ideal outcome that satisfies every person with the existing conventional image enhancement techniques. In this paper, we proposed a simple and efficient reinforcement learning based image contrast enhancement method for personal preference. Our method consists of state, action, reward or punishment definition, and policy learning. We have implemented Q-learning and State Action Reward State Action (SARSA) algorithms. The training process is easy for any user by clicking some buttons in our developed graphical user interface (GUI). The experimental results demonstrate good performance of our proposed method in this paper.

Keywords:

Image enhancement, Reinforcement learning, Linear transformation, Nonlinear transformation, Q-learning, SARSA.

1. Introduction

Image contrast enhancement is a subjective problem depending on personal preference and subject field property. Normally, image contrast enhancement is parameter dependent. Different persons or different subject fields need different parameter values. The early work for subjective contrast enhancement is based on reinforcement learning with four linear point operation actions [1]. Recently, reinforcement learning based image enhancement is receiving more and more attention [2,3,4]. The authors in [4] proposed deep Q-learning for underwater image enhancement. There exist some surveys on reinforcement learning for image enhancement Images [2, 3].

In this paper, we proposed a new and simple reinforcement learning based image contrast enhancement method by using linear and nonlinear transformations from [4, 5]. Our work in this paper is different from [1] in the following:

1) We define more actions
2) We implement both Q-learning and SARSA algorithms
3) Our source code is available in https://github.com/SemereGr/RL_contrast_enhancement

Our method is parameter free approach for image enhancement. The experimental results verify the good performance of our proposed method by subjective and objective evaluations.

1.1. Reinforcement learning

Reinforcement learning is a machine learning method where the agent depends on the feedback received from the user to update the knowledge it has about the environment. It allows the agent to learn the optimal policy to do a specific action on its environment.

Q-learning is a common reinforcement learning algorithm that seeks to find the best action to take given the current state. The learner updates the Q-table according to the following equation:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha \left[r + \gamma \left(\max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right) \right]. \quad (1)$$

Where s_t, a_t, s_{t+1}, a_{t+1}, α and γ are the current state, current action, next state, next action of the agent, learning rate and discount rate, respectively. Equation (1) can be interpreted as the agent was in state s_t, does action a_t, and changed its state to s_{t+1} from where it decides to do the action that maximizes the Q-value.

SARSA is an on-policy reinforcement learning algorithm that estimates the value of the policy being followed. The learner updates the Q-table according to the following equation:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha\big[r + \gamma\big(Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)\big)\big]. \quad (2)$$

Equation (2) can be interpreted as the agent starts in state s_t, does action a, gets reward r, moves to state s_{t+1}, decides to do an action a_{t+1}, and updates the Q-value.

Q-learning chooses the action to be performed in the next state that maximizes the Q-table while SARSA chooses the action to be performed in the next state according to probability distribution. In other words, SARSA updates the Q-table considering the action in the next state.

2. Proposed method

Our Q-learning and SARSA based image enhancement methods are summarized in Algorithm 1 and 2, respectively. The steps in each algorithm are described in more detail in the following.

Algorithm 1 image enhancement based on Q-learning

1) initialize the Q-table
2) randomly select an original image
3) compute its state based on its histogram
4) use some operation to alter the contrast of the image to get deteriorated image
5) using $\epsilon -$ greedy strategy to choose action a from possible action sets. Perform the selected action image to get an enhanced (transformed) image
6) compute the new state s_{t+1} of the transformed image
7) compare the contrast of the transformed image and deteriorated image; obtain a reward according to Table 1
8) update the Q-table based on the training tuple (s_t, a_t, s_{t+1}, r_t) from above step. Set $s_t = s_{t+1}$ and repeat step 5 to 8 until a satisfying image is retrieved
9) repeat step 2 to 8 until the Q-table converges

Algorithm 2 image enhancement based on SARSA

1) initialize the Q-table
2) randomly select an original image from the image set or training set
3) modify the randomly chosen original image to get a deteriorated image
4) determine the state of the deteriorated image s_t
5) select an action a using softmax action selection
6) perform action a on the deteriorated image to get the transformed or enhanced image
7) determine the state s_{t+1} of the transformed image
8) display the deteriorated image and the transformed image on the GUI and get feedback (rating)from the user
9) update Q-table using SARSA
10) end training when the stopping criteria is met; other- wise, perform step 2 through 10

2.1. Selection of an Original Image

The two algorithms begin with random selection of an original image from a set of images. The original image can be either a color image or a grey image. The chosen color image is transformed to grey image as this paper focuses on gray image enhancement.

2.2. Deteriorating/Modifying the Original Image

Original images are good quality images. We deteriorate the contrast of the original image to obtain different, poor contrast image using the following three transformations:

1) linear point transformation
2) non-linear transformation
3) linear and non-linear transformation

Linear point transformation is achieved by randomly choosing two numbers between [0,255] (the lowest and highest intensity values of a pixel) and the original image is scaled or squashed linearly between these values.

2.3. Determining the State of the Image

To define the state of an image, we consider the three values: minimum intensity value g_{min}, maximum intensity value g_{max}, and most frequent pixel value g_{hmax}. Each of them is discretized into three levels: low (L), middle (M), high (H), according to the following equation:

$$states = \begin{cases} L, & p \leq I_{max}/3 \\ M, & I_{max}/3 < p \leq 2I_{max}/3 \\ H, & p > 2I_{max}/3 \end{cases} \quad (3)$$

Where p is the value of g_{min}, g_{hmax}, or g_{max} and I_{max} is the maximum pixel intensity value(255). Thus, we have 10 possible image states as shown in Table 1.

Table 1. Possible states

states	g_{min}	g_{hmax}	g_{max}
s_1	L	L	L
s_2	L	L	M
s_3	L	L	H
s_4	L	M	M
s_5	L	M	H
s_6	M	H	H
s_7	M	M	M
s_8	M	M	H
s_9	M	H	H
s_{10}	H	H	H

2.4. Policy Selection

A policy is a strategy which guides an agent to select an action. In training, an agent needs to explore different possible actions and states. In this paper, we used two action selection methods:

1) $\epsilon -$ greedy action selection
2) softmax action selection

The $\epsilon -$ greedy policy is given in the following equation:

$$policy = \begin{cases} random\ action, & in\ \epsilon\ probability \\ argmax_a Q(s,a), & in\ 1-\epsilon\ probability \end{cases} \quad (4)$$

In this policy, the agent chooses the best action with $1-\epsilon$ probability and takes an action randomly, not including the best action, according to $\epsilon\ probability$. ϵ is initialized to 0.2 and decreased along iteration.

The softmax method uses a Boltzmann distribution in equation (5) to randomly choose an action at iteration t.

$$P(a|t) = \frac{e^{\frac{Q_{t-1}(a)}{\tau}}}{\sum_b e^{\frac{Q_{t-1}(b)}{\tau}}} \quad (5)$$

Where τ is computational temperature and decreased as the number of iterations increases. The higher the temperature, the more equally likely the actions are for selection. As the temperature approaches to 0, softmax action selection becomes the same as greedy action selection. τ can be decreased according to the following formula:

$$\tau = \frac{\tau_0}{\sqrt{k}} \quad (6)$$

Where k is the k^{th} iteration and τ_0 is set to 1.

2.5. Action on the deteriorated image

The actions to be performed on the deteriorated image should fix at least one of the following problems:

1) Image is too bright
2) Image is too dark
3) Image is too much contrast
4) Image is blurred

We define seven actions as below:

Action 1: Logarithmic function:

$$b \cdot ln(ax+b), \quad (7)$$

where $a = \frac{1}{40}, b = \frac{255}{\ln(255a+1)}$.

Action 2: Exponential function:

$$\frac{e^{\frac{x}{b}}-1}{a}, \quad (8)$$

where $a = \frac{1}{40}, b = \frac{255}{\ln(255a+1)}$.

Action 3: Shifted and scaled version of inverse sigmoid function:

$$255 \cdot \frac{1-\frac{\ln\left(\frac{255}{x}-1\right)}{5}}{2}. \quad (9)$$

Action 4: Shifted and scaled version of sigmoid function:

$$\frac{255}{1+e^{-5\left(\frac{2x}{255}-1\right)}}. \quad (10)$$

Action 5: Adaptive non-linear stretching using piece-wise linear transformation [5]:

$$y_i = \begin{cases} 0, & if\ x_i \le x_{min} \\ \frac{k}{b-x_{min}} \times (x_i - x_{min}), & if\ x_{min} < x_i \le b \\ k + \frac{(1-k)(x_i-b)}{x_{max}-b}, & b < x_i < x_{max} \\ 1 & x_i \ge x_{max} \end{cases} \quad (11)$$

where

$$k = \begin{cases} \mu + b(1-\mu), & k' > \mu + b(1-\mu) \\ k' & b(1-\mu) \le k' \le \mu + b(1-\mu) \\ b(1-\mu) & k' < b(1-\mu) \end{cases} \quad (12)$$

and

$$k' = b \times \frac{(1-b)\bar{y} - \int_b^1 xp(x)dx + b\int_b^1 p(x)dx}{\int_0^b xp(x)dx + b\int_b^1 p(x)dx - b\int_0^1 xp(x)dx} \quad (13)$$

$$b = mean(x) = \bar{x} = \int_0^1 xp(x)dx \quad (14)$$

where $\bar{y}$ is the predicted average value of the transformed image and is set to 0.5 in this paper.

Action 6: Min-max contrast stretching [5]:

$$y_i = \frac{x_i - x_{min}}{x_{max} - x_{min}} \quad (15)$$

where x_i and y_i are values of brightness on initial and transformed image, x_{min} and x_{max} are minimum and maximum values of brightness on initial image.

Action 7: Adaptive non-linear contrast stretching [6]:

$$\widetilde{b_i} = \tilde{b}_{low} + \left(\tilde{b}_{upp} - \tilde{b}_{low}\right) \cdot f(b_i) \quad (16)$$

$$\mathrm{f(b_i)} = \mathrm{k} \cdot \int_0^{\mathrm{b}} \varphi(0,\mathrm{z})^{\gamma}\, \varphi(\mathrm{z},1)^{\gamma}[\varphi(0,\mathrm{z})^{\gamma} + \varphi(\mathrm{z},1)^{\gamma}]^{-2}\, dz \quad (17)$$

$$k = \left(\int_0^{\mathrm{b}} \varphi(0,\mathrm{z})^{\gamma}\, \varphi(\mathrm{z},1)^{\gamma}[\varphi(0,\mathrm{z})^{\gamma} + \varphi(\mathrm{z},1)^{\gamma}]^{-2}\, dz\right)^{-1} \quad (18)$$

$$\varphi(b_1, b_2) = \int_{b_1}^{b_2} p(z)dz\,, 0 \le b_1 \le b_2 \le 1 \quad (19)$$

where $\widetilde{b_i}, b_i, \tilde{b}_{low}, \tilde{b}_{upp}$ are output intensity, input intensity, lowest intensity and maximum intensity in output image, respectively. And $p(z)$ is probability density function of pixel intensity.

Note: the first four actions can be also implemented with polynomials. for instance, a polynomial of degree 5 can be used as shown in the following:

$$f(x) = \omega_0 + \omega_1 x + \omega_2 x^2 + \omega_3 x^3 + \omega_4 x^4 + \omega_5 x^5 \quad (20)$$

where ω_0 through ω_5 are weights or polynomial coefficients.

2.6. Reward and Punishment

A graphical user interface is designed to make the feedback convenient. Two images, deteriorated image and enhanced image, are displayed on the GUI to the user so as that he/she can compare them and give his/her feedback. A mean opinion score, in the range 1-5, is used to measure the users evaluation of each modified image. The mean opinion score is translated to reward and punishment according to Table 2.

Table 2. Reward or punishment

MOS Score	Description	Reward/Punishment
1	Excellent	0.4
2	good	0.2
3	fair	0
4	poor	-0.2
5	bad	-0.4

Five buttons, namely Excellent, Good, Fair, Poor and Bad, are provided. The user only needs to click one of the buttons that expresses the quality improvement or deprecation from the deteriorated to the modified image.

2.7. Updating the Q-table

The Q-table is updated after modifying the current image and receiving feedback from the user. The Q-table will converge to the optimal Q-table (optimal policy) once the agent has explored and exploited enough. In this paper, the Q-table is updated by Equation (1) and (2).

3. Performance metrics

The following four metrics are used to evaluate the performance of the agent.

1) Histogram

A histogram is an approximate representation of the distribution of numerical data. In this paper, the histogram of the mean opinion scores is used as a metric to evaluate the agent performance.

2) Reward to punishment ratio (RPR)

The reward to punishment ratio is defined as:

$$\frac{R}{P} = \frac{number\ of\ rewards\ up\ to\ current\ iteration}{number\ of\ punishments\ up\ to\ current\ iteration} \quad (21)$$

3) Total average (TA) is defined as:

$$TA = \frac{\sum_n r_n}{N}, \quad (22)$$

where N is the maximum iteration, and r_n is the reward/punishment received at the n^{th} iteration.

4) Running average (RA)

$$RA = \frac{\sum_{i-10}^{i} r_n}{N}, \quad (23)$$

where i is the i^{th} iteration, and, r_n is the reward/punishment received at the n^{th} iteration.

4. Results

For training and test, we developed a GUI as shown in Fig 1. The user interacts with the GUI by clicking the buttons to view the images, assess the image quality, and provide rewards or punishments for the agent. Any personal or biased selections can lead to personal agent and results. In this software, we implemented both Q-learning and SARSA algorithms. The upper part shows deteriorated Lena image and enhanced Lena image by our agent.

We use this GUI to evaluate the performance of our proposed method in this paper subjectively and objectively. Images from standard digital image processing dataset are used to train our agent. These images are deteriorated through three ways: 1) linear point transformer, 2) non-linear transformation 3) hybrid of linear and non-linear transformation.

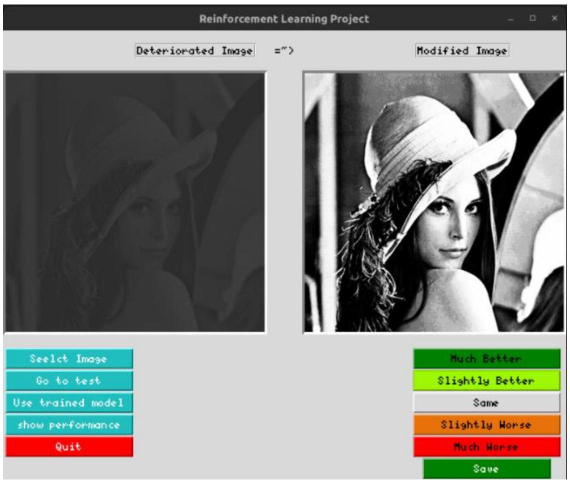

Fig.1 GUI.

Fig 2 shows values of the 4 objective performance metrics (histogram, RPR, TA, RA) in training phase. Both the first (Q-learning) and second approaches (SARSA) are trained for 48 iterations. The reward histogram and the total average (TA) for the two approaches illustrate that the two approaches got more rewards than punishments. It indicates the efficiency of agent learning method.

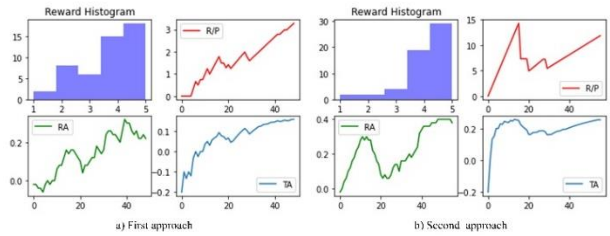

Fig.2 Metrics values in training

To subjectively show the performance of our proposed method, the contrast enhanced images are shown in Fig 3 for both algorithms. In Fig 3, The images from top row to bottom row are original good quality images, deteriorated images, results from our first method, results from our second method, respectively.

It is obvious that our two proposed approaches have enhanced greatly the deteriorated images (see last two rows in Fig 3). The green boxes indicate that the two approaches perform well in recovering the important details. The yellow boxes indicate that our method loses some details, because the cloud and the sky parts in the deteriorated image have similar pixel intensity values. The blue and red boxes indicate that the second approach slightly outperforms the first approach. However, both of the two approaches are capable of enhancing contrast deteriorated images.

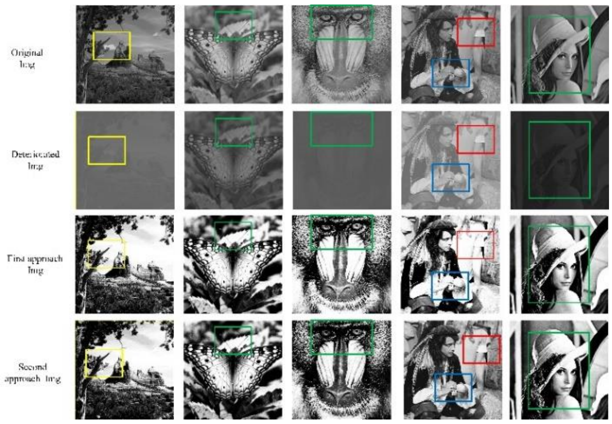

Fig.3 Contrast enhanced results through our proposed Q-learning and SARSA methods.

We show one of our personal contrast enhancement results in Fig 4, from which it can be seen that our proposed method is efficient in contrast enhancement and that some important details have been emphasized.

Fig 4. Test results.

5. Conclusion

In this paper, we proposed a new and simple reinforcement learning based image contrast enhancement method by using both linear and nonlinear transformations. We define new actions and implement both Q-learning and SARSA algorithms. Our method is parameter free approach for image enhancement. The experimental results demonstrate the good performance of our proposed method by subjective and objective evaluations.

The scope of the two proposed approaches is limited to the use of Q-learning, SARSA, ten states and seven actions and focuses on global contrast adjustment. In future work, this can be extended further to the employment of Deep Q-Network (DQN); the performance of the DQN can be improved by increasing the number of states and actions. Furthermore, both local and global contrast enhancement or other image feature enhancement (such as edge) is the potential research direction for reinforcement learning in the future.

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